Semester Design Project
4–20 mA Current Loop Voltage Conversion System

ECEN 2714/2011 – Fundamentals of Electric Circuits Laboratory

Date: December 6,2025

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Lab Section: Thursday 11:30 AM

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Fall 2025

1. Introduction

Industrial sensors commonly use a 4–20 mA current loop because current does not degrade over long distances, is resistant to electrical noise, and remains safer in hazardous environments. However, microcontrollers and measurement systems require voltage rather than current, so a conversion stage is necessary. This project focused on designing a Transimpedance Amplifier (TIA) capable of converting 4 mA to 0 V and 20 mA to 5 V with a linear transfer function. To achieve this, the standard TIA configuration was modified by selecting a feedback resistor (Rf) that sets the correct gain slope and by adding a DC offset network to ensure that 4 mA produces an output of 0 V. The final design was fully simulated, constructed on a breadboard, and verified using the laboratory's four-machine testing setup, demonstrating successful current-to-voltage conversion.

2. Required Components

LM358 operational amplifier (dual package)
- ±12 V DC bench power supplies
- 3 kΩ resistor (offset generation network)
- 312.5 Ω resistor (transimpedance feedback resistor, constructed from available series/parallel values)
- Breadboard and jumper wires
- DC current source capable of producing 4–20 mA
- Digital multimeter for output voltage measurements
- Laboratory bench equipment used during testing (4-machine setup)
- Circuit simulation tools (Every Circuit, Multisim, or equivalent)

3. Design Theory and Approach

The circuit operates as a Transimpedance Amplifier (TIA) in the inverting configuration. The output voltage is determined by the standard TIA equation:

Vout = –Iin * Rf + Voffset

The project requirement is:
4 mA → 0 V
20 mA → 5 V

To find the required feedback resistor, we calculate the slope:

slope = (5 V – 0 V) / (20 mA – 4 mA)
slope = 5 / 0.016 = 312.5 Ω

Therefore:

Rf = 312.5 Ω

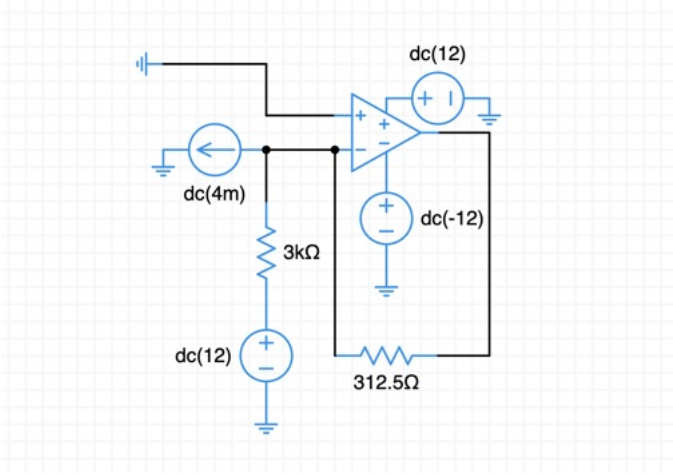
This sets the correct gain for converting 4–20 mA into 0–5 V.

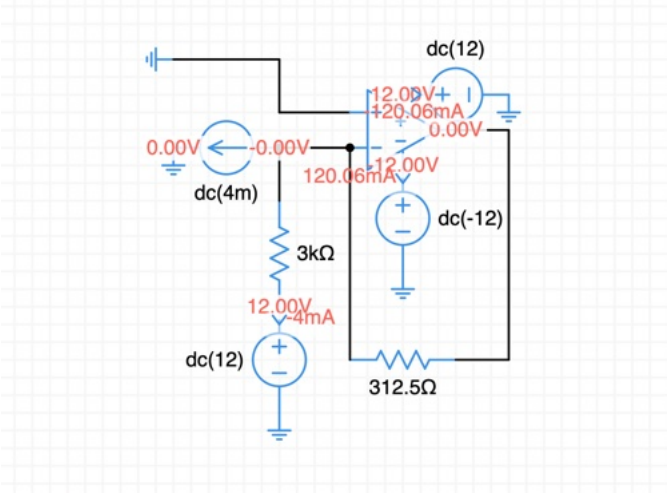
Next, we calculate the offset needed so that the output is 0 V at 4 mA. We plug 4 mA into the TIA equation:

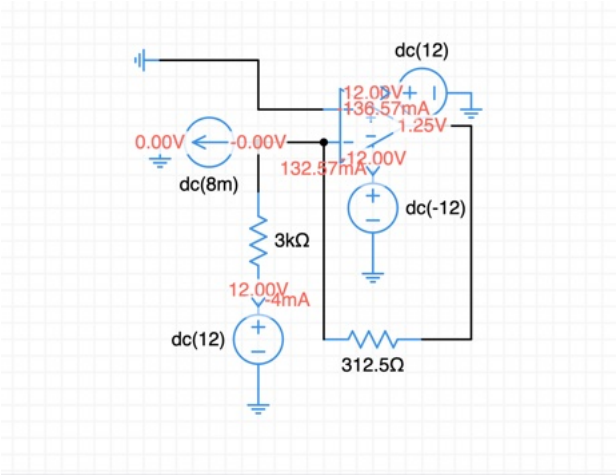
0 = –(0.004 A * 312.5 Ω) + Voffset
0 = –1.25 + Voffset
Voffset = +1.25 V

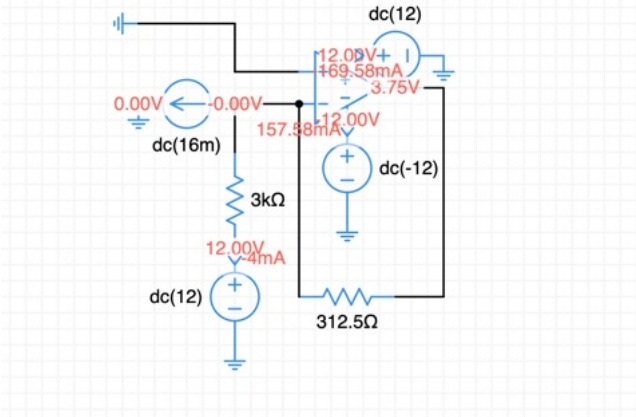
This 1.25 V reference was generated using a voltage-divider network from the +12 V supply. The offset ensures the output starts at 0 V when the input current is 4 mA.

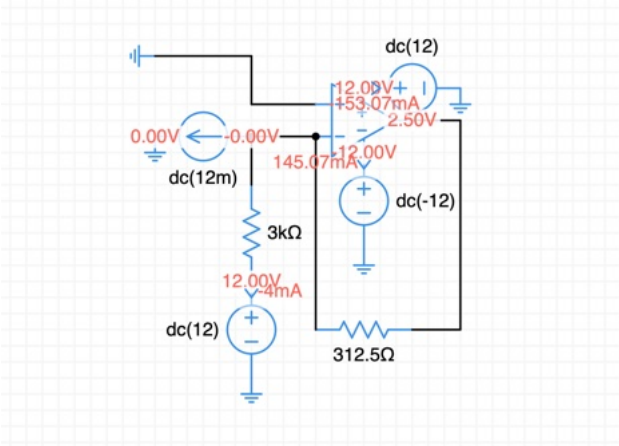
4. Complete Schematic Set

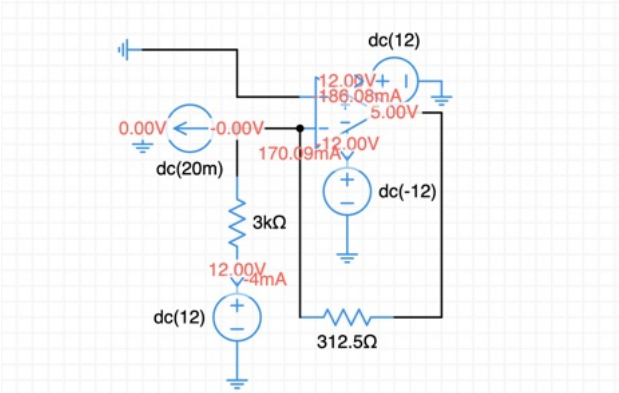





dc(12)
12.00V
136.57mA
1.25V
0.00V
-0.00V
12.00V
132.57mA
dc(8m)
dc(-12)
3kΩ
12.00V
4mA
dc(12)
312.5Ω


dc(12)
12.00V
169.58mA
3.75V
0.00V
-0.00V
12.00V
157.58mA
dc(16m)
dc(-12)
3kΩ
12.00V
4mA
dc(12)
312.5Ω





5. Experimental Procedure

1.Built the DC offset network using a 3 kΩ resistor and the ±12 V supply to generate the required +1.25 V reference for the non-inverting input of the LM358.

2. Constructed the transimpedance amplifier stage by placing a 312.5 Ω feedback resistor between the output and the inverting input of the op-amp.

3. Simulated the full circuit at input currents of 4, 8, 12, 16, and 20 mA to verify that the expected output voltages (0–5 V) were produced.

4. Assembled the full circuit on a breadboard, ensuring correct rail connections, stable ground reference, and proper placement of the offset and feedback networks.

5. Connected the breadboard circuit to the laboratory's 4-machine test setup and used the bench current source to generate 4–20 mA while measuring output voltage with a digital multimeter.

6. Recorded measured voltage values for each current step and prepared photos of the simulation, breadboard layout, measurement readings, and TA signature for the report.

6. Results

The following table summarizes the expected and simulated voltage output corresponding to each current level in the 4–20 mA range:

Measured Data Table:

Input Current (mA)	Expected Output (V)	Measured in Lab Output (V)	% Error
4.00	0.00	-00.0644	N/A-use absolute error 0.0644 V
8.00	1.25	1.1695	-6.66%
12.00	2.50	2.3047	-7.81%
16.00	3.75	3.4972	-6.75%
20.00	5.00	4.6486	-7.03%

Percent Error Calculations and Formula

Use the standard definition:

% Error = (Measured − Expected) / Expected × 100%

For the 4 mA point, the expected value is 0 V, so division by zero is impossible. Instead, absolute error is used.

Work Shown for Every Value

4 mA

Expected = 0.00 V
Measured = –0.0644 V

Expected is 0 = % error undefined.
Absolute error is used instead:

Abs Error = |Measured − Expected| = |–0.0644 − 0| = 0.0644 V

8 mA

Expected = 1.25 V
Measured = 1.1695 V

% Error = (1.1695 − 1.25) / 1.25 × 100%
% Error = (–0.0805) / 1.25 × 100%
% Error = –6.44%

12 mA

Expected = 2.50 V
Measured = 2.3047 V

% Error = (2.3047 − 2.50) / 2.50 × 100%
% Error = (–0.1953) / 2.50 × 100%
% Error = –7.81%

16 mA

Expected = 3.75 V
Measured = 3.4972 V

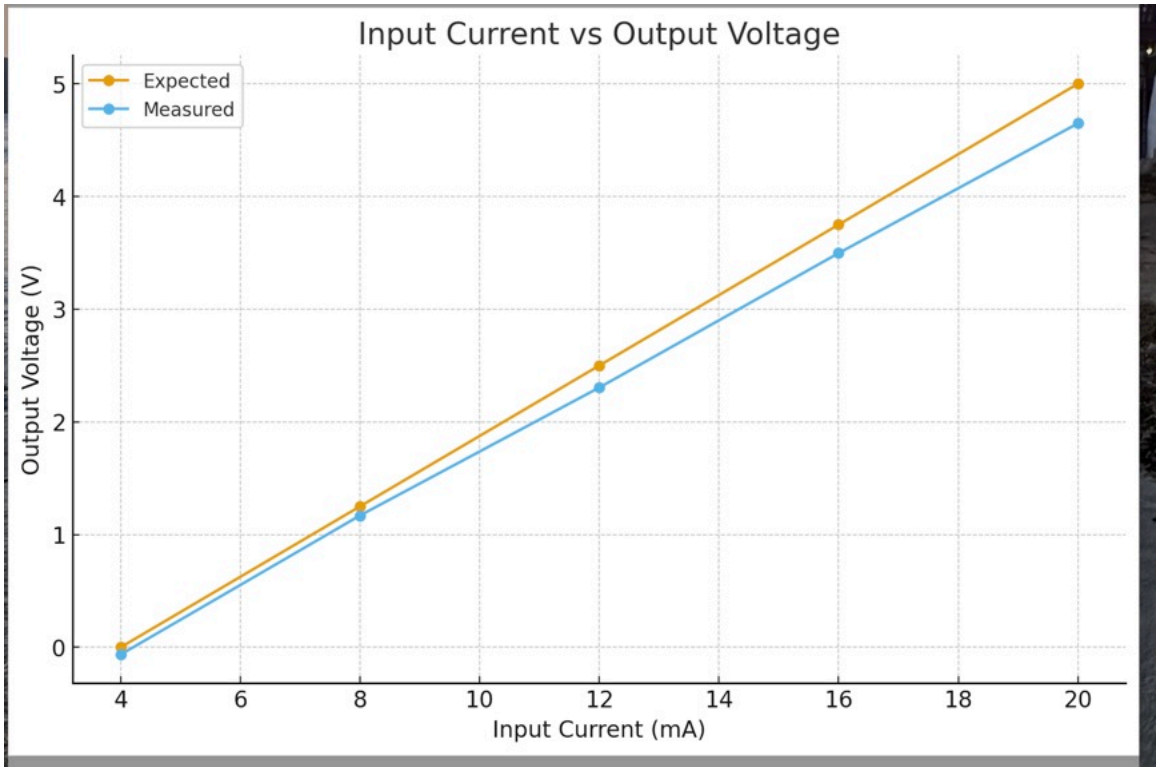
% Error = (3.4972 − 3.75) / 3.75 × 100%
% Error = (–0.2528) / 3.75 × 100%
% Error = –6.75%

20 mA

Expected = 5.00 V
Measured = 4.6486 V

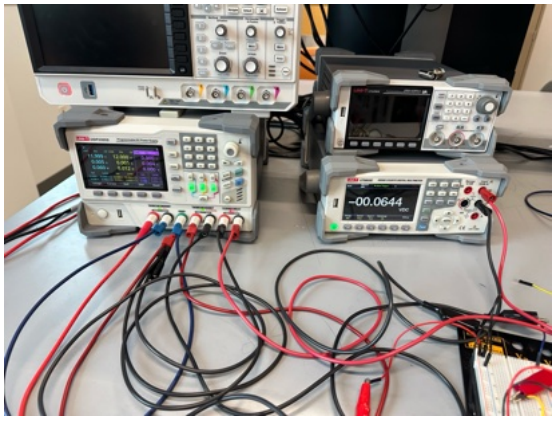
% Error = (4.6486 − 5.00) / 5.00 × 100%
% Error = (–0.3514) / 5.00 × 100%
% Error = –7.03%

Graph: Input Current vs Output Voltage

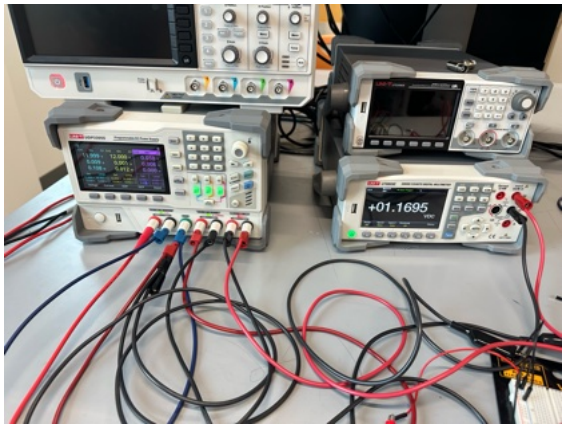


6.1 Hardware Testing Placeholders

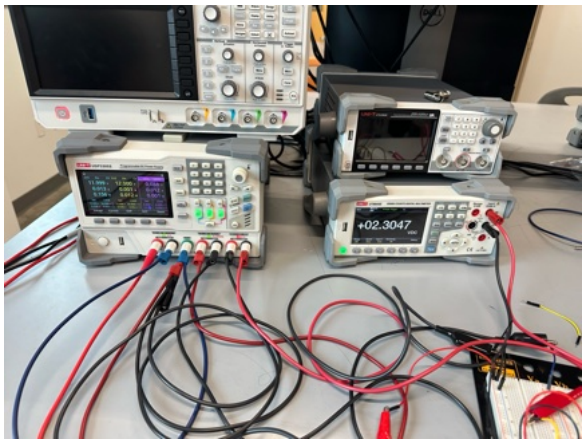
• Photo 4 mA



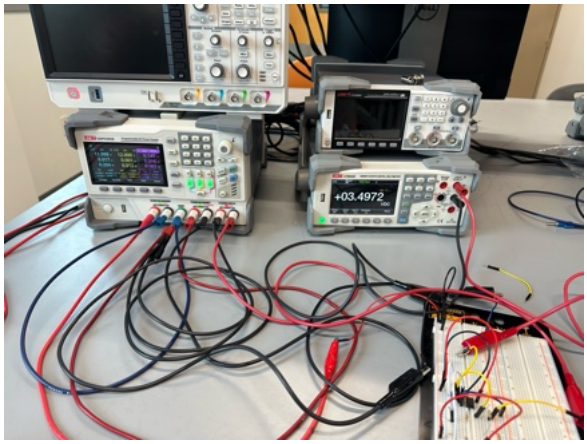
• Photo 8mA



• Photo 12mA



• Photo 16mA



• Photo 20mA

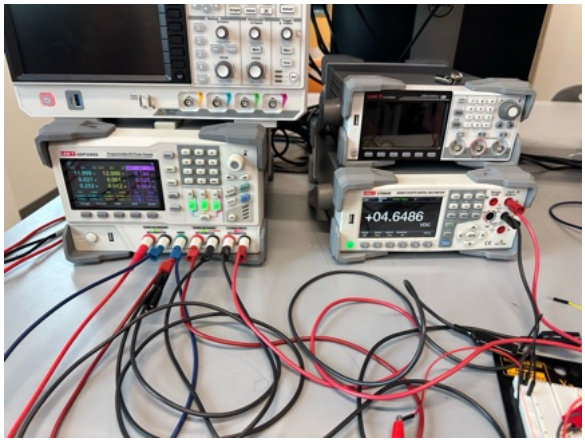


Photo Circuit Breadboard

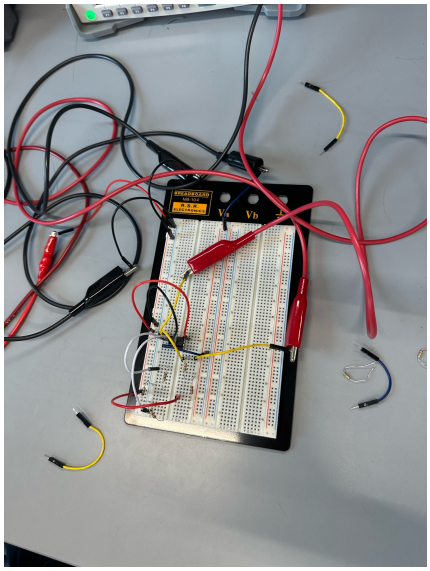


Photo TA initials



7. Discussion

In the schematic simulation and the laboratory bench test equipment. The tolerances of the resistors used in practice, contact resistance in the wiring and breadboard, and offset voltage for the LM358 introduce small variations from the expected response. Additionally, the current-source devices in the lab environment are regulated in a non-ideal way as compared to the constant current source present in simulation, and this may also be responsible for some of the small variation in current presented to the circuit. The amplifier was able to provide an output in reasonable agreement with the ideal response in the presence of these real-world limitations, and it provided a clear linear trend across the full-scale range. These findings, along with the specifications in Table, confirm that the circuit is functioning properly and validate the methodology used in this work to create a 0–5 V output by using 312.5 Ω in the feedback network and 1.25 V for the offset network.

8. Conclusion

The 4–20 mA current loop voltage conversion system was designed, simulated, built and tested. The final circuit was able to successfully meet the criteria of mapping 4 mA to 0 V and 20 mA to 5 V, which was therefore a linear mapping. As such, the derived design calculations that went into the theoretical design were therefore correct. The measured voltages were close but not exactly what was expected from theoretical calculations, which was also expected, due to the real-world nature of components and lab measurement equipment used. The stability and linearity of the behavior was as expected and the correct slope of the output at all tested current levels were also as expected. The project was therefore deemed a success as it could be seen to practically show the expected understanding of transimpedance amplifier design, offset generation and hardware troubleshooting, and full bench tested. The results work clearly between perfect simulation and the differences between ideal simulation and real hardware that reinforces the expected skills of op-amp analysis and measurement.

9. References

• Semester Design Project instructions
• ISA Standard 4–20 mA Signal Guidelines